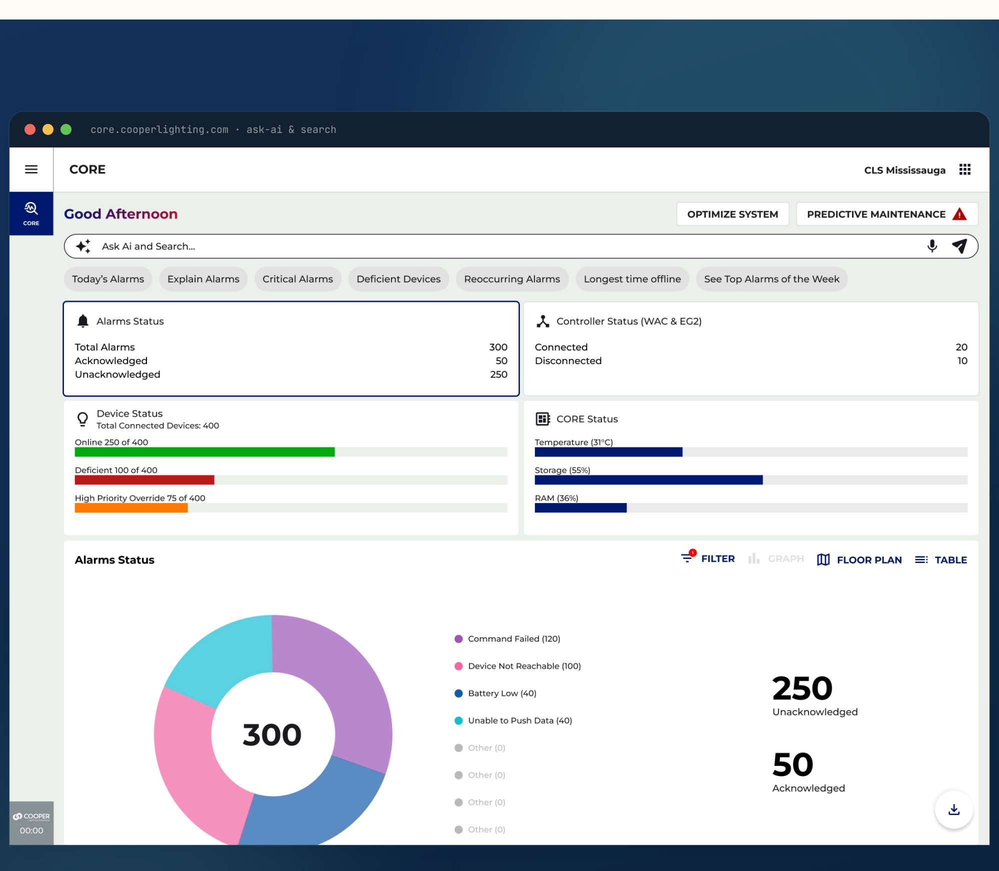
ENTERPRISE AI · IOT PLATFORM · 2026

Infusing intelligence into the *enterprise*

Embedding conversational and predictive AI into CORE, Cooper Lighting's cloud platform for managing connected lighting across enterprise buildings. Turning a wall of cryptic alarms into plain-language answers, and reactive maintenance into foresight.

Read the full case study → See the three pillars



CORE DASHBOARD, ASK AI & SEARCH, ONE-TAP INTENTS, ALARM AND DEVICE ROLL-UP

THE BRIEF

From a wall of alarms to *a conversation*

CORE manages thousands of connected lighting devices across enterprise buildings, controllers, sensors, wall stations, relay packs. At scale, the platform generated a relentless stream of alarms and status codes that only specialists could decode, and maintenance stayed stubbornly reactive: teams found out a device had failed only after it went dark.

My mandate was to define how AI should live inside an established enterprise product, not a bolted-on chatbot, but woven through the daily workflow. Three pillars emerged: conversational access, explainability for every alarm, and prediction ahead of failure.

ROLE Lead Experience Designer	SCOPE AI strategy → delivery Interaction & content
PLATFORM CORE · Web Cloud	DOMAIN Connected lighting · IoT
OPERATORS Facility manager · Field tech CLS sales agent	YEARS 2026

THE PROBLEM, IN NUMBERS

Signal that only *specialists could read*

A single large site carried hundreds of devices and a daily flood of fault codes, almost none of which a facility manager could decode without help. AI had an obvious job from day one.

01
Alarm overload
Up to 300 alarms a day at peak across a 400+ device site, 14+ distinct fault codes, few of them decodable without a specialist.

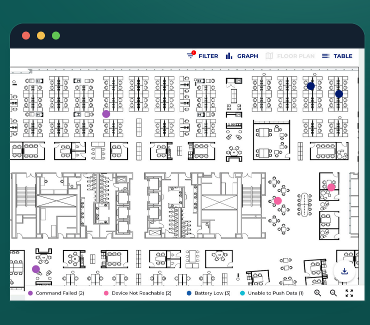
02
Three different reads
A facility manager, a field tech on a ladder, and a sales specifier all need the same intelligence, framed for very different jobs.

03
Reactive, never ahead
Maintenance only happened after a device went dark, the system never warned anyone a failure was coming.

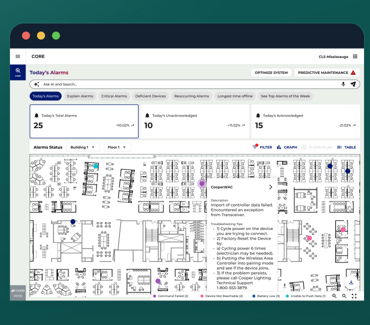
CONVERSATIONAL AI

One bar to *ask the building anything*

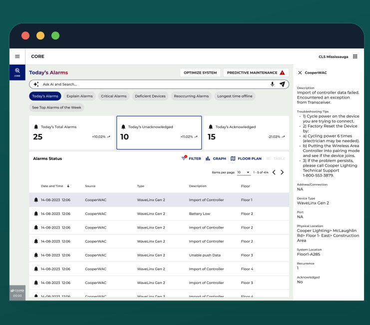
A single Ask AI & Search bar sits atop every screen, with one-tap intents for the questions teams ask each morning. The same answer reads three ways, a floor plan, a building roll-up, or a table.



FLOOR PLAN, ALARMS AS PINS ON THE LIVE MAP



BY BUILDING, PORTFOLIO ROLL-UP



TABLE, TRIAGE AT VOLUME

APPROACH

Explanation, *plus a next step*

AI only earns its place if it helps the person on the floor at 2am. Framing the assistant as explanation plus a concrete action, not open chat, kept every answer specific, sourced from a real device, and trustworthy.

01 **Intents, not syntax**
One-tap chips, Today's Alarms, Critical Alarms, Deficient Devices, Longest Time Offline, encode the questions teams already ask, so nobody learns a query language.

02 **Every code, translated**
Explain Alarms turns any cryptic fault code into a plain-language cause and a numbered remediation, the answer a senior specialist would give, available to anyone.

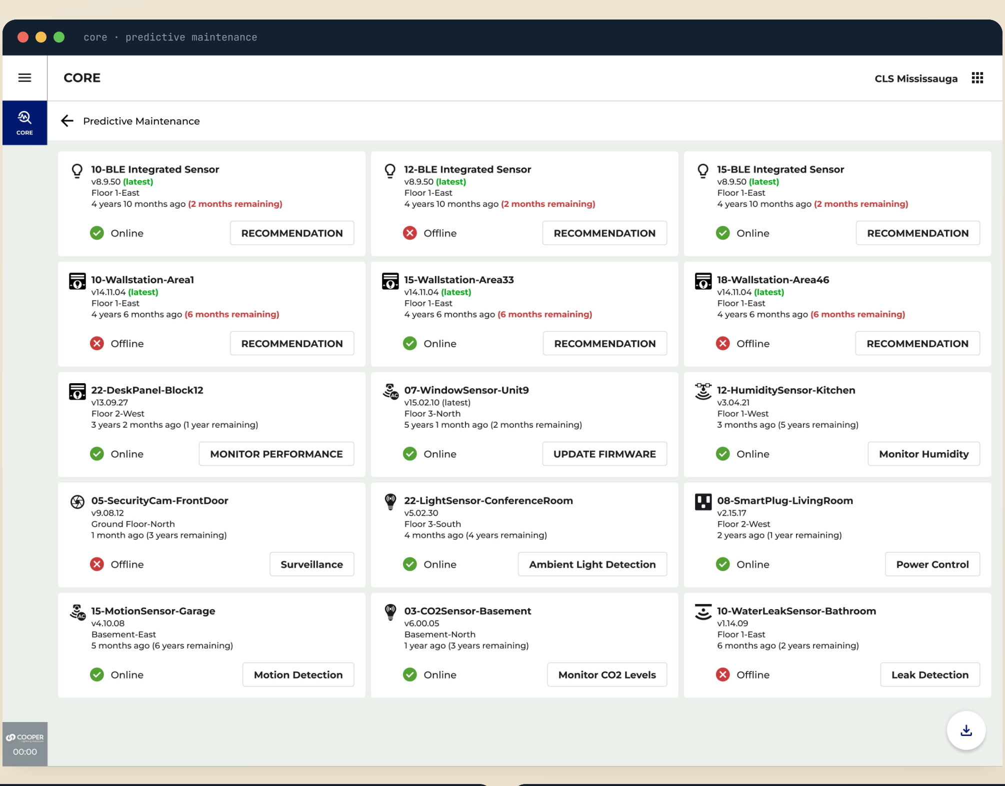
03 **Fix it before it goes dark**
Predictive Maintenance shows firmware, install age, and an AI-estimated lifespan, surfacing devices roughly two months before end-of-life with a one-tap recommendation.

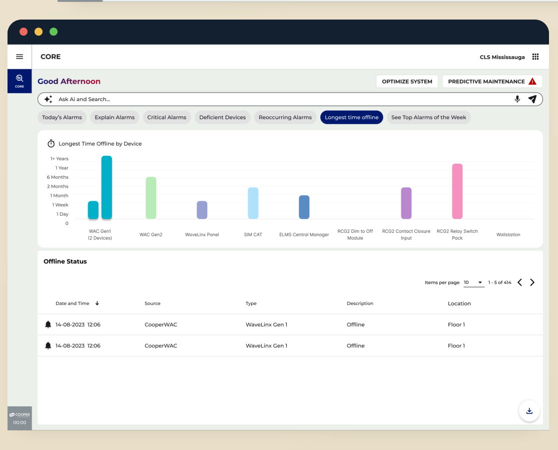
04 **Set the intent, let it tune**
Cooper Intelligence takes a single goal, energy or comfort, and tunes occupancy, dimming, and schedules across the site, then keeps learning from real use.

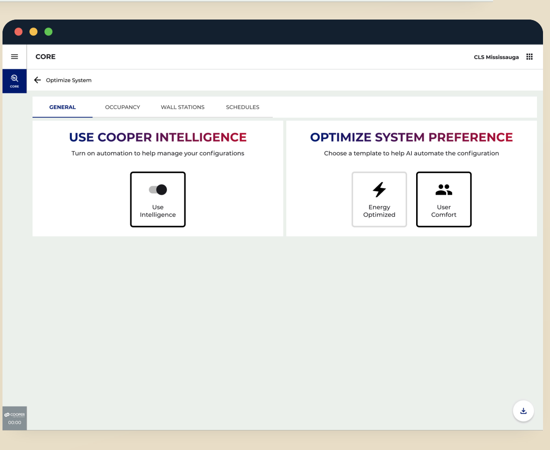
EXPLAINABILITY & PREDICTION

Every alarm, *translated to a plan*

The shipped Explain Alarms panel pairs a plain-language cause with a numbered fix, while Predictive Maintenance ranks devices by silence and estimated lifespan so the worst actors surface first.





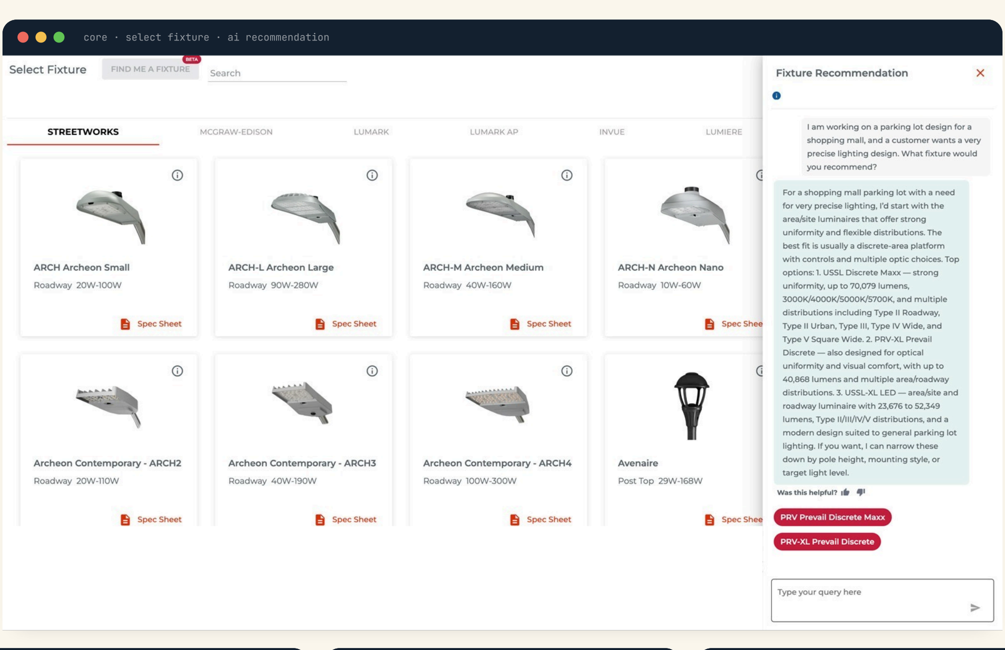


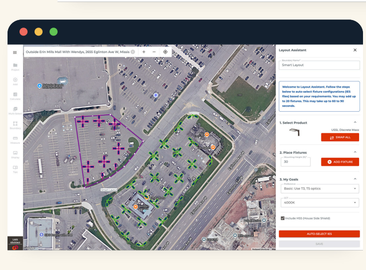
LONGEST TIME OFFLINE RANKING · OPTIMIZE SYSTEM, COOPER INTELLIGENCE TEMPLATES

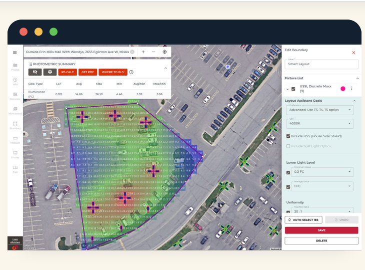
AI IN THE FIELD

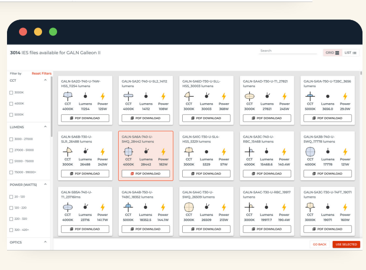
From a brief to *the right fixture*

The same intelligence reaches sales and specification. A conversational Fixture Recommendation assistant turns a plain brief into specific products, then connects to a searchable library of 5,000 IES files and an Area Wizard that auto-places fixtures on a real site map.







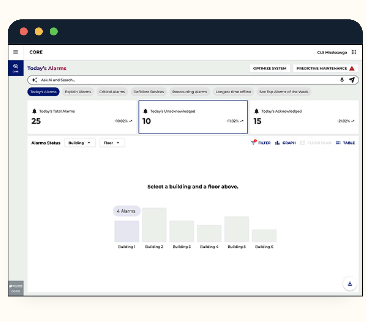


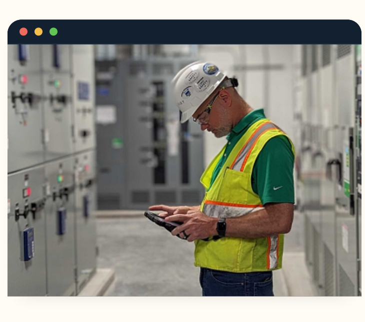
Searchable IES library, 5,000 photometric files filtered and exported in a click · Smart Layout Area Wizard auto-places up to 20 fixtures to hit a target level and uniformity.

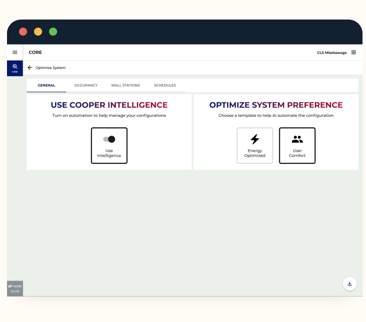
OUTCOME

Built for the people *holding the pager*

Anchoring AI to existing pain, the alarm wall, gave it an obvious job from day one. The next step is making confidence and provenance first-class, showing why the AI predicted a failure, not just that it did, and exposing what the optimization loop has learned over time.







BY THE NUMBERS

What intelligence bought

3
Pillars: conversational, explainable, predictive
CONVERSATIONAL · EXPLAINABLE · PREDICTIVE

14+
Alarm codes turned into plain-language remediations
ALARM CODES · 14+ TRANSLATED
ONE DOT PER CODE · 14 DRAWN

5000
IES photometric files made searchable
IES LIBRARY · 5,000 FILES
FULL LIBRARY SEARCHABLE · ONE-CLICK EXPORT

20
Fixtures auto-placed per area by Smart Layout
AUTO-PLACED · UP TO 20
ONE DOT PER FIXTURE

SHIPPED CORE AI WORK · COOPER LIGHTING · 2026.

> *The goal was never to add AI to a dashboard. It was to make a 400-device building answer a question, and tell you what to do next, the way the best technician on the team would.*